LAP-TEQ PLUS Elevation — User Manual

TEQSAS PRODUCTS

LAP-TEQ PLUS Elevation

User Manual

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1 Before you begin

1.1 Intended use

The LAP-TEQ PLUS ELEVATION sensor is used to determine the correct mounting height of loudspeakers. The LAP-TEQ ELEVATION sensor measures the vertical distance between the device and the surface located directly below it. The measured distance is shown on a separately purchased LAP-TEQ PLUS Display, LAP-TEQ PLUS Interface or a LAP-TEQ Motion System. Any other use that is not described in these instructions for use is unauthorised misuse. The manufacturer is not liable for any damage resulting from this. Never use the device in public traffic (road traffic, aviation, etc.).

1.2 What do the symbols used mean

Hazard warnings and notes are clearly marked in the instructions for use. The following symbols are used:

DANGER

Immediate danger to life or injury!

Immediately dangerous situation that will result in death or serious injury.

WARNING

Risk of death or injury!

Generally dangerous situation that may result in death or serious injury.

CAUTION

Potential risk of injury!

Dangerous situation that may result in injury.

NOTICE

Risk of damage to the appliance!

Situation that may result in material damage.

> NOTE
>
> Information provided for a better understanding of the processes.

2 For your safety

WARNING!

Failure to observe the safety information and instructions may result in electric shock, fire and/or serious injury!

Read all safety information and instructions.

2.1 General safety instructions

- Keep all safety information and instructions for future reference. The term “appliance” used in the safety instructions refers to mains-powered appliances (with mains cable) and battery-powered appliances (without mains cable).
- Danger to life and accidents for infants and children! Never leave children unattended with packaging material and product. There is a risk of suffocation from packaging material and a risk to life from strangulation. Children often underestimate the dangers. Always keep children away from the product. The product is not a toy.
- To ensure safe handling of this appliance, the user must have read and understood these instructions for use before using the appliance for the first time.
- If you sell or pass on the appliance, it is essential that you also hand over these instructions for use.
- This appliance is not intended for use by persons (including children) with reduced physical, sensory or mental capabilities, or lack of experience and/or knowledge, unless they have been given supervision or instruction concerning use of the appliance by a person responsible for their safety.
- The appliance may only be used if it is in perfect working order and fully assembled. If the appliance or any part of it is defective, it must be taken out of service and repaired or disposed of properly.
- Use the appliance only for the purpose for which it is intended.
- Keep children away from the appliance! Keep the device safe from children and unauthorised persons.
- Only use and store the device within the permissible ambient conditions (temperatures, humidity, etc.).

2.2 Handling laser beams

RISK OF EYE INJURIES FROM LASER BEAMS

If the laser beam enters the eyes through optical instruments, irreparable damage may occur.

Never look into the laser or aim the laser beam at reflective surfaces!

- Do not look into the beam, even from a great distance.
- Never aim the measuring beam at people, other living beings or reflective surfaces.
- Standard laser goggles do not protect against the dangers of laser radiation. They merely make the laser beam easier to see.

2.3 Workplace safety

- Keep your work area clean and well lit. Untidiness or unlit work areas can lead to accidents. Observe the workplace and accident prevention regulations applicable in your country.
- Do not work with the electrical appliance in potentially explosive atmospheres containing flammable liquids, gases or dust. Power tools generate sparks that can ignite the dust or vapours.

2.4 Electrical safety

- Static electricity can occur in dry environments. In dry rooms, touch a metal object to discharge the static electricity before operating the appliance.
- Do not misuse the cable to carry or hang up the appliance or to pull the plug out of the socket. Keep cables away from heat, oil, sharp edges or moving parts of the appliance. Damaged or tangled cables increase the risk of damage to the appliance.
- If you use the appliance outdoors, only use extension cables that are also suitable for outdoor use. The use of an extension cable suitable for outdoor use reduces the risk of damage to the appliance.

2.5 Handling strong magnets

- Magnets can affect the function of pacemakers and implanted defibrillators. As a wearer of one of these devices, you must not stay in the immediate vicinity!
- The holding force of magnets can decrease. Always reckon with a magnet coming loose and the device falling down. Therefore, the device and mounting bracket must always be secured with safety ropes during use!
- Magnets have a strong attractive force. When handling them, be careful not to pinch your fingers or skin. Wear safety gloves if necessary.
- Magnets are brittle and can splinter. Avoid magnets impacting each other or hard surfaces. Do not work them mechanically by drilling or grinding.

- Magnets and their coatings can contain nickel. Regular contact with them can lead to the development of nickel allergies. Avoid touching the magnets with your skin if possible. If you have an existing nickel allergy, allergic reactions may occur. In this case, refrain from handling magnets.
- Magnets can irreversibly damage objects through their magnetic field. Maintain a sufficient safety distance of at least 1 metre. Items that may be damaged include, for example:
  - Electronic devices such as hearing aids, laptops and monitors
  - Iron-containing objects such as mechanical watches
  - Data carriers such as magnetic strips on credit cards, hard drives, etc.

2.6 Safety of persons

- Be attentive, pay attention to what you are doing and use common sense when working. Do not use the device if you are tired or under the influence of drugs, alcohol or medication. A moment of inattention during use can lead to serious injuries (e.g. by electric shock).

CAUTION

Risk of injury!

Improper repairs can result in your appliance no longer functioning safely. You are putting yourself and your surroundings at risk

NOTICE

Risk of damage to the appliance!

All components are maintenance-free. Never open the devices, otherwise the warranty will be invalidated. An exception is opening the lower cover to set the measurement unit and the correction mode.

2.7 Service

- Only have your appliance repaired by qualified specialists and only with original spare parts. This will ensure that the safety of the power tool is maintained.
- Ensure adequate lighting when working with the appliance. Poor visibility can lead to an increased risk of accidents.
- Do not use any accessories that have not been specially designed and recommended by the manufacturer for this appliance. Just because you can connect the accessories to your appliance does not guarantee trouble-free and safe use.
- The device may show signs of wear and tear over time.

- The use of unauthorised accessories may cause damage or wear that is not covered by the warranty.

2.8 Special device instructions

- Do not use the appliance in potentially explosive atmospheres or in the vicinity of flammable liquids or gases!
- Paint and stickers can stick to moving parts and impair proper operation.
- If you have an allergic reaction to the colour or metallic parts of the appliance, it may cause itching, eczema or swelling of the skin. If this is the case, stop using the appliance and consult a doctor.
- Broken glass or plastic can cause injury. Have the appliance repaired at the service centre.
- Do not drop the device and do not subject it to shocks.
- The device or parts of it may be damaged if dropped, bent or deformed.
- The device has safety precautions that limit the leaked output of the laser beam: no changes may be made to the optical system.
- This device is classified as a Class 2 laser product in accordance with the international standard IEC 60825. It works with a visible laser beam that is harmless under normal operating conditions.
- Symbols on your device must not be removed or covered. Labels on the appliance that are no longer legible must be replaced immediately.

NOTE

Read and observe the instructions for use before commissioning.

RISK OF EYE DAMAGE!

Never look directly into the laser beam.

3 Product Description

3.1 Scope of delivery

The scope of delivery includes the following parts:

- 1x LAP-TEQ PLUS ELEVATION sensor
- 1x Safety rope
- 1x Manfrotto 5/8” sleeve (for mounting on the MOUNTING BRACKET)
- Instructions for use and safety data sheet

3.2 Your device at a glance

Figure 1: LAP-TEQ PLUS Elevation – Overview

1. Laser radiation outlet
2. Receiving lens
3. XLR socket for connecting a LAP-TEQ PORT Compatible Sensor
4. XLR socket for connection to the readout unit (LAP-TEQ PORT)
5. Mounting pin for attachment to the LAP-TEQ PLUS MOUNTING BRACKET
6. Safety rope

> NOTE
>
> In the factory setting, the trigonometric correction is activated and the unit is set to “m” (metres). To deactivate the trigonometric correction and/or to change the unit to “ft” (feet), please proceed as described in the chapter “Trigonometric Correction / Length Unit” on page 6.

The LAP-TEQ PLUS ELEVATION module is intended for mounting on a flying frame (using the LAP-TEQ PLUS MOUNTING BRACKET) and measures the vertical distance between the zero line on the device (see chapter Your device at a glance on page 9) and the surface located below it. This provides information about the height of a loudspeaker element or of the entire array. The laser beam of the ELEVATION serves as visual feedback for the area up to which measurement is taken. This way you can see whether the end point of the measurement is correct. A LAP-TEQ PLUS Sensor can additionally be attached to the MOUNTING BRACKET. This allows simultaneous measurement of height and vertical alignment. Both values can be read on the display at the same time. If a LAP-TEQ MOTION System is used as the display unit, height and alignment can be adjusted simultaneously.

3.3 Optional accessories

The following accessories can be purchased for operation of the LAP-TEQ ELEVATION:

- LAP-TEQ PLUS MOUNTING BRACKET
- Transport case
- …

4 Powering up

4.1 Connection

NOTICE!

Risk of damage to the appliance!

Moisture and liquid can cause damage to parts or electronic circuits inside the device.

- Do not switch on the device when it is damp.
- If the device is already switched on, switch it off.
- If the device cannot be switched off, leave it in this state to dry
- Wipe the device dry with a dry cloth and bring it to a service centre.

NOTICE!

Risk of damage to the appliance!

All components are maintenance-free. Never open the devices, otherwise the warranty will be invalidated. An exception is opening the lower cover to set the measurement unit and the correction mode

Figure 2: Connecting ELEVATION module to display module

NOTE

The cable length (2) between DISPLAY and ELEVATION must not exceed 70 m. If you want to bridge distances greater than 70 m, an XLR cable with a cross-section larger than 0.22 mm^2 must be used.

- Connect the ELEVATION module (1) at the “IN” input and the display module (3) using a standard 3-pin XLR cable (2).

Any number of ELEVATION units can be connected to a single display module in succession. For example, three ELEVATION units can be used to verify correct alignment of the left and right PA sides as well as a centre cluster.

For correct operation, the display module must be connected to an ELEVATION via a standard 3-pin XLR connection cable. The ELEVATION module is powered via the XLR cable. There are no rechargeable or non-rechargeable batteries inside the ELEVATION.

> NOTE
>
> The ELEVATION module (1) is equipped with Neutrik ® XLR TOP connectors. The IP65 protection of these connectors can only be guaranteed if a connector of the XLR TOP series is also installed on the connection cable (2).

4.2 Connection of a LAP-TEQ PLUS Sensor (optional)

Figure 3: Connecting LAP-TEQ PLUS Sensor with ELEVATION module

- Mount the LAP-TEQ PLUS Sensor (1) likewise on the flying frame or on the MOUNTING BRACKET (see instructions for use of the LAP-TEQ PLUS).
- Connect the LAP-TEQ PLUS Sensor (1) to the “THRU” socket of the ELEVATION module (3) using a standard 3-pin XLR cable (2).

> NOTE
>
> The total length of the cables (2) + (4) between LAP-TEQ PLUS Sensor, ELEVATION and display module must not exceed 45 m. If you want to bridge distances greater than 45 m, an XLR cable with a cross-section larger than 0.22 mm^2 must be used.

4.3 Mounting the MOUNTING BRACKET (optional)

- Place the magnetic holder on the upper edge of the flying frame.
- Secure the magnetic holder to the flying frame against falling using the safety rope
- Route the XLR cable in a loop through the strain relief on the magnetic holder.
- Check that the ELEVATION module can swing freely.
- Make sure that there are no objects in the measurement path. These can lead to incorrect measurements

5 Usage

5.1 Switching on/off

> NOTE
>
> Make sure that the reference point of the measurement is correct. If the deviation from the vertical is so great that the laser hits an object, for example, this will lead to an incorrect measurement result.

Check that the device is in a safe condition:

- Check whether there are any visible defects.
- Check that all parts of the device are firmly attached.

> RISK OF EYE INJURIES FROM LASER BEAMS
>
> If the laser beam enters the eyes through optical instruments, irreparable damage may occur.
>
> Never look into the laser or aim the laser beam at reflective surfaces!

- Make sure that no persons are in the effective range of the laser.
- Switching on: press the “Power” button on the display module for approx. 3 seconds.

If the system has been put into operation correctly, the red laser lights up and the current distance is shown on the display. If a LAP-TEQ PLUS Sensor is also connected, the current angle reading is also shown.

The reading is updated continuously; under unfavourable conditions (cf. Observe ambient conditions on page 25), the update may take up to 3 seconds.

- Switching off: press the “Power button” on the display module again.

6 Cleaning

DANGER!

Danger to life from electric shock!

Disconnect the XLR cable to the display module (power supply) before carrying out any work on the device.

NOTICE!

Risk of damage to the appliance!

Do not immerse the device in water when cleaning, as water can penetrate the electrical system and damage it.

NOTICE!

Possible product damage!

The surface can be damaged. Do not use scratching or abrasive objects or aggressive cleaning agents for cleaning.

- Use a soft, lint-free cleaning cloth dampened with mild soapy water to clean the housing.
- Avoid scrubbing motions on the display window and on the glass window covering the laser outlet
- Do not use chemicals, cleaning agents or solvents. This can lead to discolouration and corrosion of the outside of the device or result in an electric shock or fire.

7 Issues and Help

> CAUTION!
>
> Risk of injury!
>
> Improper repairs can result in your appliance no longer functioning safely. You are putting yourself and your surroundings at risk

It is often only minor faults that lead to a malfunction. In most cases, you can easily rectify these yourself. Please refer to the following table before contacting your dealer. This will save you a lot of trouble and possibly also costs.

If the device requires servicing, take it to your sales partner.

Error/Fault	Cause	Remedy
Device does not work	Device defective?	Contact service
Laser not visible	Poor ambient conditions?	See note on ambient conditions in this manual
	Cable too long?	Use a QLX cable with a larger cross-section
	Measurement distance too long?	If possible, reduce the measurement distance
“?.?m” or “?.? ft” appears on the display	The device cannot find an end point for the measurement	Check the measurement end point and/or the line of sight. Can the ELEVATION module swing freely? Place reflective material (paper) on the measurement end point.
Three dashes appear on the display	XLR cable not correctly plugged in	Push the XLR cable in firmly

If you cannot rectify the fault yourself, please contact our service team directly. Please note that improper repairs will also invalidate the warranty and you may incur additional costs.

7.1 Service centre

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TEQSAS GmbH
Otto-Hahn-Straße 20a
50354 Hürth
Deutschland
Tel.: +49 (0)2233 611-500
E-Mail: service@teqsas.de
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7.2 Observe ambient conditions

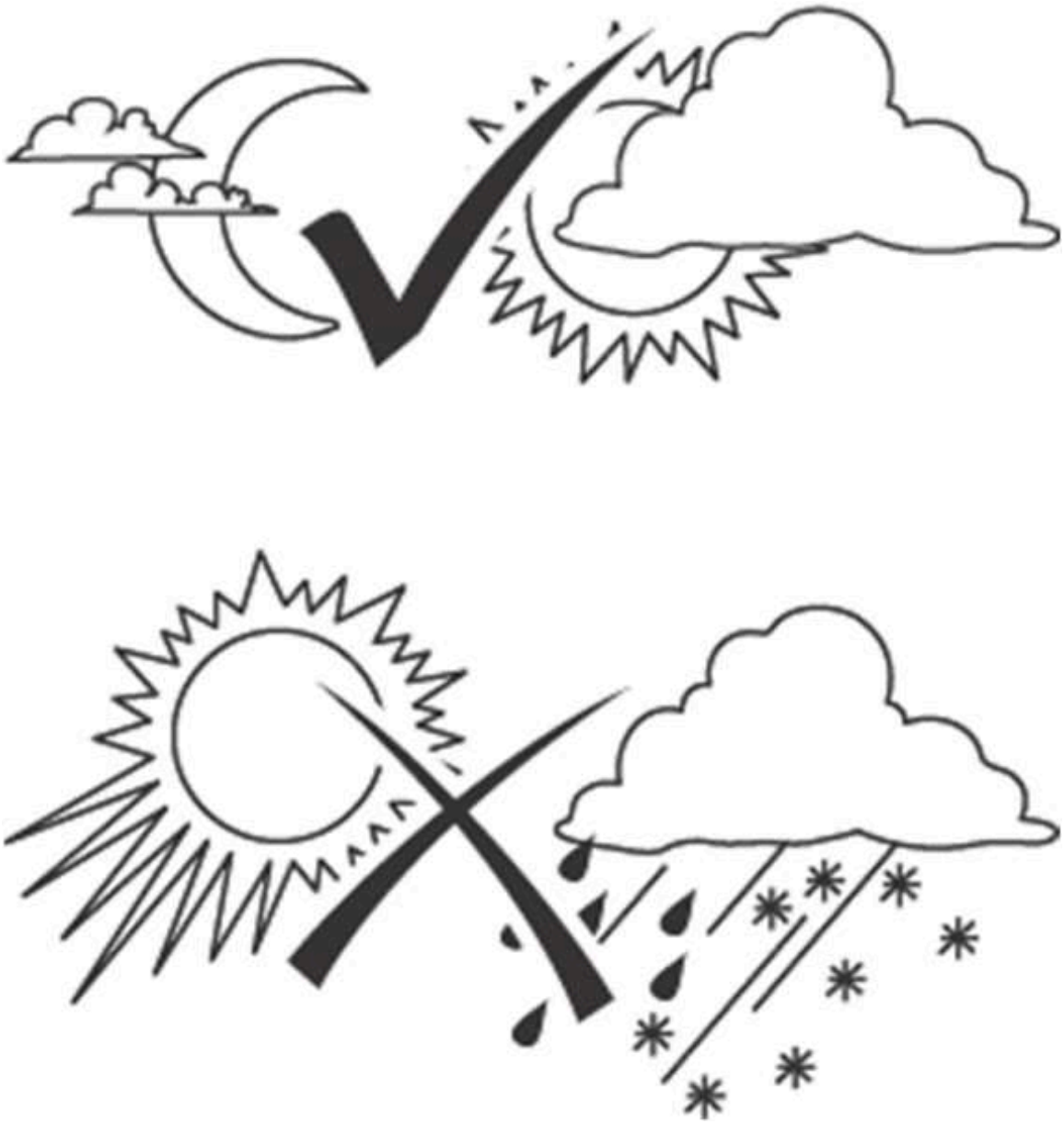

Figure 4: Ambient conditions / laser beam visibility

The visibility of the laser beam depends on the ambient brightness. As a general rule:

- Subdued light = good visibility
- Direct sunlight = poor visibility
- Rain, dust or steam = poor visibility

Other adverse influences:

- Measurements through glass or plastic panes
- Contaminated laser outlet
- Strong temperature fluctuations: always allow the device to acclimatise in the transport case for some time before use in very cold or warm environments.

8 Trigonometric Correction / Length Unit

> NOTE
>
> In the factory setting, the length unit is set to “m”. It can be changed to “ft” (feet).

> NOTE
>
> In the factory setting, the trigonometric correction is activated.

If the ELEVATION module cannot swing freely (e.g. due to cable tension), the trigonometric correction can correct the deviation.

If the trigonometric correction is activated, a deviation from the vertical is corrected. The corrected value appears on the display. If it is deactivated, the actually measured – and possibly incorrect – value of the oblique measurement is displayed

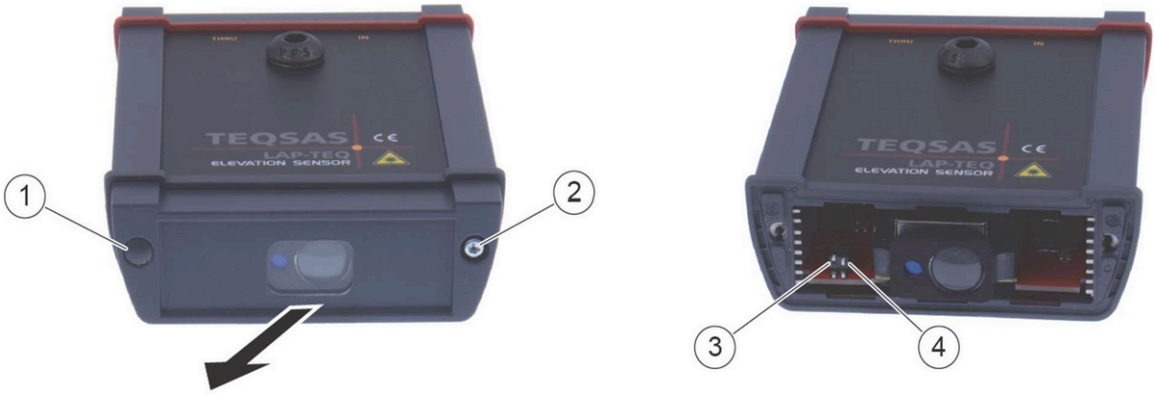

Figure 5: DIP switches on the display module

- Remove both cover caps (1) on the display module using a small screwdriver
- Unscrew both screws (2)
- Pull off the cover
- Set the DIP switches (3) and (4)
- Set the length unit to metres or feet – DIP switch 1, left (3):
  - Metres: set DIP switch 1 to “OFF”
  - Feet: set DIP switch 1 to “ON”
- Activate/deactivate trigonometric correction – DIP switch 2, right (4):
  - Activate trigonometric correction: set DIP switch 2 to “ON”
  - Deactivate trigonometric correction: set DIP switch 2 to “OFF”
- Replace the cover
- Tighten both screws (2)

- Press both cover caps (1) back on

9 Storage

- Clean and dry all parts and store them in a clean and dry place.
- If you do not use the device for a long time or are transporting it, use the original packaging.

NOTICE!

Risk of damage to the appliance!

High temperatures can damage the device or permanently reduce the holding force of the magnets.

Do not store the device in very hot places such as the interior of a car in summer or places exposed to direct sunlight.

10 Disposal

The packaging is made of environmentally friendly materials that can be disposed of at local recycling centres.

Devices marked with the symbol shown alongside must not be disposed of with household waste. You are obliged to dispose of waste electrical and electronic equipment separately.

You can find out how to dispose of old appliances from your local authority.

By disposing of old appliances separately, you are sending them for recycling or other forms of reuse. In doing so, you will help to prevent harmful substances from being released into the environment.

10.1 Disposing of batteries

Batteries that are no longer needed do not belong in household waste — they must be disposed of in accordance with regulations.

- Take used batteries to a battery collection point at the dealer or municipality
- Do not burn batteries. Risk of explosion!
- Do not burn rechargeable batteries. Risk of explosion!

11 Technical Data

LAP-TEQ PLUS ELEVATION

LAP-TEQ PLUS ELEVATION	
Protection rating	IP40
Optics	Spot optics
Laser class	2(<1 mW)
Laser colour	635nm (red)
Measuring range	0.05m to 40m (under optimal conditions)
Measuring tolerance range	+/-1mm (under optimal
Permissible ambient temperature for use	0 – 40°C
Permissible ambient temperature for storage	–20 – 50°C
Dimensions	130 × 99 × 108mm
Weight ELEVATION	650g
Order number	1670

The manufacturer reserves the right to make technical changes to the product / product group without prior notice.

EU Declaration of Conformity

Figure 6: CE

in accordance with EU Directive 2014/30/EU (Electromagnetic Compatibility) of 26 February 2014 and 2014/35/EU (Low Voltage Directive) of 26 February 2014.

We hereby declare that the device designated below, in its design and construction and in the version placed on the market by us, complies with the essential health and safety requirements of EU Directive 2014/30/EU and the requirements for electromagnetic compatibility. This declaration shall lose its validity if the device is modified without our agreement.

Manufacturer

TEQSAS GmbH
Otto-Hahn-Straße 20a
50354 Hürth

The manufacturer bears sole responsibility for issuing this declaration of conformity.

Description of the device:
LAP-TEQ PLUS ELEVATION

and its identical or similar derivatives.

Place and date of issue: Hürth